

NITIN YADAV

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LeetCode: https://leetcode.com/u/nitin_y2807

EDUCATION

VIT Bhopal University

B.Tech in Computer Science Engineering | CGPA: 8.03 / 10

Bhopal, Madhya Pradesh

Final Year | Expected 2027

M.L.P. Senior Secondary School

Class XII | 79.97%

Rewari, Haryana

Euro International School

Class X | 85.8%

Rewari, Haryana

TECHNICAL SKILLS

AI / LLM: Gemini API, Prompt Engineering, LLM Applications, RAG, Vector Search, Sentence Transformers, ChromaDB

Languages: Python, SQL, TypeScript, C++

Backend & APIs: FastAPI, Flask, REST APIs, Pydantic, JWT Auth, SQLAlchemy, Alembic, PostgreSQL, SQLite

Frontend & Tools: React, Vite, JavaScript, HTML/CSS, Git, GitHub, Docker

AI / ML: Scikit-learn, TensorFlow, CNNs, Grad-CAM

PROJECTS

Repix: AI-Powered Codebase Assistant

Python, FastAPI, Gemini, ChromaDB, React

GitHub: <https://github.com/Nitin-2807/Repix>

- Built an end-to-end **RAG application** that lets developers ask natural-language questions about any GitHub repository and get **grounded, repository-aware answers** from **Gemini**.
- Engineered the ingestion pipeline: clone repo, parse files, chunk code, embed with **Sentence Transformers (all-MiniLM-L6-v2)** and index in **ChromaDB** for semantic retrieval.
- Tuned the retrieval and prompting flow to cut irrelevant context, keeping responses tied to the actual codebase.
- Developed **FastAPI/Pydantic** REST APIs with **PostgreSQL, SQLAlchemy, Alembic** and **JWT auth**, plus a **React + Vite** chat interface.

Crop Disease Detection using AI

TensorFlow, CNN, Grad-CAM, Flask, SQLite

GitHub: <https://github.com/Nitin-2807/Crop-Disease-Detection>

- Built a custom **CNN with SeparableConv2D layers** in TensorFlow to classify **15 crop-disease classes** from 128×128 leaf images, reaching **79.97% accuracy** on the PlantVillage dataset.
- Developed the end-to-end image preprocessing, training and evaluation pipeline, and packaged the trained model into a **Flask web app** backed by **SQLite**.
- Integrated **Grad-CAM explainability** to visualize which regions of a leaf image drive each prediction, making the model's decisions interpretable.
- Used Grad-CAM heatmaps to inspect model behaviour on **PlantDoc real-world field images** and study how background, lighting and image conditions influence predictions.
- Combined **deep learning, computer vision and explainable AI** into an application aimed at practical agricultural use.

ACHIEVEMENTS

- Solved **300+ DSA problems on LeetCode** (graphs, dynamic programming, trees, shortest paths, greedy).
- Built working AI applications at **AI-focused hackathons**, taking ideas from prototype to functional product under time pressure.